

Operating Systems and Networking

Network Programming

IDENTIFICATION

CODE : IF-4-PR
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 8.0 h
Laboratory : 8.0 h
Project : 0.0 h
Teacher-student
contact : 25.0 h
Personal work : 30.0 h
Total : 55.0 h

ASSESSMENT METHOD

Exam:
- Written exam
- Allowed document: one sided A4
of lecture notes

Lab evaluation:
- Type of evaluation: demonstration
- Requested data: report + source of
code of the implemented software

TEACHING AIDS

- Lecture slides
- Code for lab work

TEACHING LANGUAGE

English

CONTACT

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AIMS

The objective of this course is to introduce the paradigms and techniques that underly communication networks. The main objectives are as follows:

- Introduce the OSI model of networks
- Describe the high-level layers of the OSI model
- Present the mechanisms for network programming, for the application programmer point of view (which API, how to use them)

CONTENT

Lectures [Cours Magistraux - CM] :

- CM1 : Introduction
- CM2 : Socket-based network programming
- CM3 : RPC/RMI-based network programming
- CM4 : Application protocols (HTTP, FTP, SMTP)
- CM5 : Architecture design
- CM6 : Architecture design (cont.)

Lab [Travaux Pratiques - TP] :

- TP1 : Socket-based network programming
- TP2 : Programming an HTTP server

BIBLIOGRAPHY

- A. S. Tanenbaum, D. J. Wetherall. Computer Networks. Pearson, 2010.
- G. Pujolles. Les Réseaux

PRE-REQUISITE

Programming skills (Java, or C)